myMove: Car Parking System with Sensors

SAABIRESH LETCHUMANAN

BACHELOR OF COMPUTER SCIENCE WITH HONOURS

2025

UNIMY

UNIVERSITY MALAYSIA OF COMPUTER SCIENCE & ENGINEERING

School Of Computing and Digital Technology

myMove: Car Parking System with Sensors

SAABIRESH LETCHUMANAN

2025

myMove: Car Parking System with Sensors

SAABIRESH LETCHUMANAN

A thesis submitted
in fulfillment of the requirements for the degree of
BACHELOR OF COMPUTER SCIENCE WITH HONOURS

School Of Computing and Digital Technology

UNIVERSITY MALAYSIA OF COMPUTER SCIENCE AND ENGINEERING

2025

DECLARATION

I declare that this thesis entitled “myMove: Car Parking System with Sensors“is the result of my own research, except as cited in the references. The thesis has not been accepted forany degree and is not concurrently submitted in candidature of any other degree.

Signature: 

Name: SAABIRESH LETCHUMANMAN

Date: 12 November 2025

APPROVAL

I hereby declare that I have read this thesis and in my opinion this thesis is sufficient in terms of scope and quality for the award of Degree BACHELOR OF COMPUTER SCIENCE WITH HONOURS

Signature: ________________________

Supervisor: MUHAMMAD SYAHMIE SHABARUDIN

Date: 29 December 2025

DEDICATION

To my beloved family, including my wife and parents, I would like to thank them for their continuous love, encouragement, and support throughout my academic journey. Their patience and understanding have been my greatest source of strength during challenging times.

I would also like to dedicate this work to my supervisor and lecturers, whose guidance, advice, and constructive feedback have played an important role in shaping this project. Finally, this thesis is dedicated to all friends who offered help, motivation, and support throughout the completion of this study.

ABSTRACT

Parking challenges such as limited space availability, double parking, and inefficient parking management systems are common problems faced by drivers in busy urban areas in Malaysia. Most existing parking applications focus mainly on digital payments and often lack real-time parking information, parking reservation features, and effective communication tools to manage double-parking situations. To address these issues, this project proposes myMove, a smart mobile-based parking platform designed to improve the overall parking experience through a centralized and user-friendly system. The application focuses on features such as viewing parking information, reserving parking spaces, digital payments, booking management, and QR code–based communication between drivers without exposing personal contact details. This study was carried out during the FYP1 phase and involved a literature review and a pilot survey to identify user needs and limitations of existing systems. The findings indicate strong user interest in an integrated parking management solution and provide a solid foundation for the development, implementation, and evaluation of the full system in FYP2.

myMove: Sistem Parkir Kereta dengan Sensor

ABSTRAK

Masalah parkir seperti kekurangan tempat letak kereta, parkir berganda, dan sistem pengurusan parkir yang tidak cekap merupakan isu yang sering dihadapi oleh pemandu di kawasan bandar yang sibuk di Malaysia. Kebanyakan aplikasi parkir sedia ada lebih tertumpu kepada pembayaran digital dan sering kekurangan maklumat parkir masa nyata, ciri tempahan parkir, serta alat komunikasi yang berkesan untuk mengurus situasi parkir berganda.Bagi menangani masalah ini, projek ini mencadangkan myMove, iaitu sebuah platform parkir pintar berasaskan aplikasi mudah alih yang direka untuk meningkatkan pengalaman parkir secara keseluruhan melalui sistem yang berpusat dan mesra pengguna. Aplikasi ini memfokuskan kepada ciri-ciri seperti paparan maklumat parkir, tempahan ruang parkir, pembayaran digital, pengurusan tempahan, serta komunikasi berasaskan kod QR antara pemandu tanpa mendedahkan maklumat peribadi.Kajian ini dijalankan pada fasa FYP1 dan melibatkan kajian literatur serta tinjauan perintis bagi mengenal pasti keperluan pengguna dan kekangan sistem sedia ada. Dapatan kajian menunjukkan minat yang tinggi terhadap penyelesaian pengurusan parkir yang bersepadu dan menyediakan asas yang kukuh untuk pembangunan, pelaksanaan, dan penilaian sistem penuh dalam fasa FYP2.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor for the continuous guidance, valuable advice, and constructive feedback provided throughout the completion of this Final Year Project. Their support and encouragement were essential in helping me stay focused and motivated during the research and documentation process. I would also like to thank the lecturers and academic staff for sharing their knowledge and providing helpful insights that contributed to the development of this project. Special thanks are extended to all participants who took part in the pilot study survey, as their time, responses, and honest feedback were invaluable in identifying real-world parking challenges and shaping the requirements of the proposed system. Finally, I would like to express my deepest appreciation to my wife, my family, and my friends for their constant support, patience, and encouragement throughout this journey. Their understanding and motivation played a significant role in helping me complete this project successfully.

TABLE OF CONTENT

CHAPTER 3

CHAPTER 4

CHAPTER 5

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

Abbreviation	Full Term
FYP	Final Year Project
UI	User Interface
UX	User Experience
QR	Quick Response
GPS	Global Positioning System
IoT	Internet of Things
LPR	License Plate Recognition
VoIP	Voice Over Internet Protocol
FCM	Firebase Cloud Messaging
API	Application Programming Interface
DBKL	Dewan Bandaraya Kuala Lumpur
IC	Identify Card
ESP32	Espressif ESP32 Microcontroller

CHAPTER 1

INTRODUCTION

1.1 Background

These recent years, searching for parking has become increasingly stressful because people who own vehicles are rising day by day (Chen et al., 2025). High density areas such as condominiums, shopping malls and office or university campuses face these issues endlessly. Facilities like visitor parking, resident parking and loading bays must operate efficiently to ensure smooth traffic flow and reduce congestion (Alshareef., 2024). However, issues such as double parking, difficulty finding parking spots, and long waiting times often cause frustration for drivers. Looking at today's technologies, many individuals must have their own mobile phone that allows them to access the internet or download numerous applications (Khalid et al., 2024).

Despite improvements in today's tech, many parking areas still depend on manual checking, simple ticketing systems, and using physical cards that do not provide real-time information (Puspitasari et al ., 2021). These systems often lack important functions like digital reservations, live availability updates, and cashless payments. As a result, people spend unnecessary time searching for parking in the same area, leading to waste of time and stress. To address these challenges, the most important aspect of this project is to create a user-friendly mobile application named myMove that allows users to check availability parking, reserve parking, make digital payment, and manage double-parking situation through QR Code and in-app communication that will satisfy customer's needs.

1.2 Problem Statement

The approach to urban parking management these days encounters several significant difficulties in today's rapidly evolving cities. Drivers struggle to locate available parking spaces due to outdated systems that do not provide real-time availability or clear guidance on where to park (Kalašová et al., 2021). For example, a typical parking area in Malaysia where it lets drivers to go through a mess, like circling the same place around just to find an unoccupied parking spot. When drivers keep searching for a parking spot even though none are available, it causes stress and wastes their time, which can reduce their overall productivity. Besides, many parking systems in Malaysia are outdated and cannot keep up with the growing number of vehicles. Because they do not provide real-time information, drivers spend more time searching for parking, which increases traffic congestion and reduces the overall quality of urban mobility.

Also, double-parking remains a persistent issue as well in Malaysia, especially in crowded areas, where blocked vehicle owners have no fast or secure way to contact the driver who parked behind them (Haslinda et al., 2016). Despite DBKL’s implementation to clamp cars that park in illegal parking areas in 2016, recent studies show that the number of compounds issued has increased dramatically between January to May 2025, which DBKL also received 1,009 public complaints related to illegal parking (Anis, 2025). This shows that current enforcement methods alone are not enough to solve the ongoing illegal and double-parking issues.

1.3 Research Question

This study aims to answer the following research questions:

i. How can a mobile application improve the way drivers find and reserve parking spaces without relying on physical hardware?
ii. How does QR-based communication help drivers handle double-parking situations?
iii. Can real-time applications reduce search time for drivers in busy urban areas?
iv. Can a dedicated application like myMove effectively solve the problems in real world situations and existing platforms?

1.4 Research Objective

The aim of this project is to develop a parking system that provides real-time information, finds parking spaces easily, reserves a spot and communicate during double-parking situations to reduce stress, search time and congestion.

The objectives of the project are as follows:

i. To study existing parking systems, identify current issues faced by Malaysian drivers, analyze the limitations of current car parking application, define user requirement specification and features through literature review and survey
ii. To evaluate how the app reduces search time, stress and communication problems commonly faced by drivers.
iii. To evaluate the application usability by conducting testing with real individuals and collecting feedback for further improvement.

1.5 Scope of Research

The scope of this research is as follows:

i. Development of mobile application (Android) using Flutter and backend with Firebase for authentication.

ii. This application targets Malaysian drivers who had overwhelming and frustrating experience during their daily search for parking or dealing with double-park situation.

iii. Users will be able to create an account, which prompts them to enter details such as name, age, email, password, car number plate

1.6 Thesis Outline

Based on the objectives previously presented and, on the approach proposed before, this thesis is made up of five (5) chapters, which contents are summarized as follows:

- Chapter 1 (Introduction)
  - This chapter presents an overview of the project, including the background of the study, problem statement, objectives, scopes, contributions and significance of the research.

- Chapter 2 (Literature Review)
  - This chapter provides a review of existing literature related to the project and mobile parking applications. This chapter examines the current applications such as Touch 'n Go eWallet, JomParking and Flexi Parking.

- Chapter 3 (Methodology)
  - This chapter explains the research methodology applied in this study. It outlines the research approach, data collection methods and the software development process that will be applied for this project.

- Chapter 4 (Result and Discussion)
    - This chapter presents the results and findings obtained from the pilot study and prototype development of this project. It includes survey results, user feedback and analysis related to current parking issues and user expectations.

- Chapter 5 (Conclusion and Recommendation for Future Search)
    - This chapter concludes the study by summarizing the main findings and contributions of the research. It also discusses the limitations of the current study and provides recommendations for future work, particularly for the full system development and evaluation in FYP2.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Nowadays, technology is advancing significantly in every aspect of life and people are keeping up with current development (Aditya Kumar 2022,). In this chapter it reviews existing research, studies and technologies related to mobile applications used for parking management, real-time parking detection and driver communication. The review identifies the current challenges faced by Malaysian drivers, including double parking, outdated parking system / application and lack of real-time parking availability. It examines current solutions like Touch ‘n Go eWallet’s in-app parking, JomParking and Flexi Parking to identify their strength. This review establishes the need for a more modern and integrated parking management system that offers features such as real-time parking availability, parking reservation, digital payments, automatic number plate detection, in-app communication between drivers, push notifications for car-blocking alerts, navigation to available parking lots, and a QR-based contact system for emergencies. Together, these features aim to improve driver convenience, reduce parking-related conflicts and modernize the overall parking experience in Malaysia.

2.2 Research of Related Studies

To better understand the current landscape of mobile based parking management and communication systems, three relevant studies and existing solutions were reviewed. These studies helped identify key design features and gaps that inform the development of myMove.

TABLE 2.1 COMPARATIVE STUDY OF CAR PARKING SYSTEM

Source	Main Findings	Methodology	Strengths	Weaknesses	Relevance to Project
P. Sampath, et al. IoT- based Smart Parking System for Industry 4.0 with QR code access and real- time navigation (2025)	The study shows that using IoT sensors, QR codes, and a mobile app can reduce parking search time, congestion, and user frustration, while improving parking efficiency and security.	IoT sensors, QR codes, and a mobile app can make parking faster, safer, and less stressful.	Offers real-time parking info, QR-based access, reservations, navigation, and a working hardware-software prototype.	The system is more suitable for controlled environments like malls or campuses but difficult to scale without infrastructure support.	This study highlights myMove's key features: real-time parking info, reservations, QR-based interaction, and mobile app use.
Mahat, S. N. & Nadzir, N. M. "Accessing User Satisfaction of Smart Parking Payment System in Selangor" (2025)	User satisfaction in parking apps mainly depends on how easy they are to learn and how clearly information is shown, making usability more important than having many complex features.	A survey conducted among local users with statistical analysis to identify the factors that influence their satisfaction.	Uses data from Malaysian users to provide clear insights and recommendations on user experience.	Focuses on payment features rather than driver-to-driver communication or QR code sticker interactions.	Supports the importance of having a simple user experience, clear information, and features adapted in MyMove.
Balqis Lim "Uni students develop smart system to curb double parking problems (ParkKing)" (2017)	A smart outdoor parking system aimed at reducing double-parking by detecting illegally parked vehicles and enabling more efficient parking supervision	The project developed a ParkKing prototype using sensors to detect and manage double-parking in outdoor lots.	The system helps reduce double-parking by identifying problem parking and using technology to support enforcement and reduce traffic issues.	It lacks technical details such as system accuracy, implementation scope, real-world testing, and driver communication, focusing more on the announcement than evaluation.	This study shows a real-world effort to address double parking and highlights the need for better parking systems. It supports myMove's use of technology to manage illegal parking and shows the potential of smart detection systems in parking solutions.

2.3 Study of Existing Related Systems

This section explains about the functionality and limitations of existing platforms used by Malaysian drivers. Below are three applications, Touch 'n Go eWallet's parking, JomParking and Flexi Parking were chosen because of their popularity of the application.

2.3.1 Touch 'n Go eWallet's parking

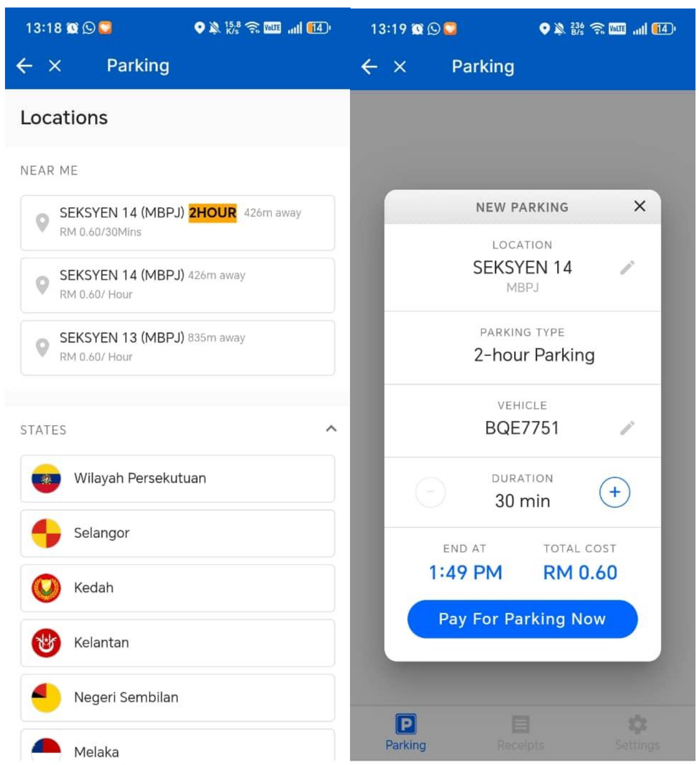

Figure 2.1: Example of Parking System in Touch 'n Go eWallet.

Touch 'n Go eWallet parking is a feature inside the Touch 'n Go eWallet app that allows users to pay for parking digitally only at selected locations / street parking. The app is convenient because there is no need to use physical tokens, it automatically deducts parking fees from their eWallet balance and has LPR Parking feature. However, there are certain features that have limitations and lacks features. It does not show real-time parking availability, so drivers still need to drive around to find an empty spot.

2.3.2 JomParking

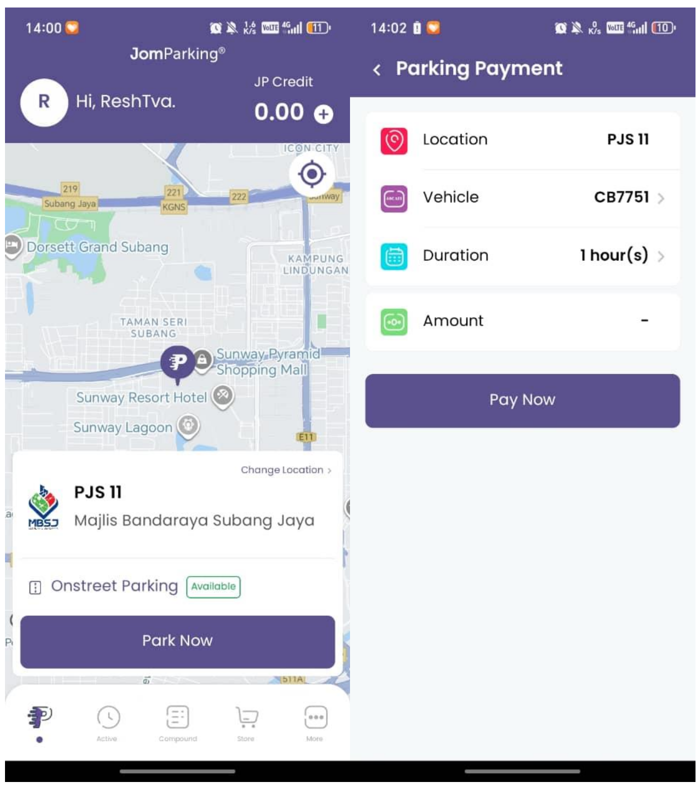

Figure 2.2: Example in JomParking

JomParking is a mobile parking application widely used in Malaysia as Touch 'n Go eWallet to pay for street and city council parking. The app allows users to select their location by dragging the map, entering their car number plate and selecting the duration of the parking time. It also provides city council compound, where users can pay for outstanding payments and includes parking passes, reload credit to the app and services like car battery delivery, booking a trip and car and motorbike insurance. However, their platform is basic and offers limited interaction. It doesn't support every city council parking around Malaysia and doesn't include every off-street parking like mall or buildings.

2.3.3 Flexi Parking

Figure 2.3: Example of Flexi Parking

Flexi Parking is a local platform in Malaysia that offers to pay for street parking under supported city councils. Users can select their parking zone, add durations, their car number plate and pay using in-app credit. However, it's not fully operational and optimized in terms of parking experience. The platform sometimes runs into an error, like connection issues, payment and top out problems, and receiving suspicious messages about traffic fines.

2.4 Suggested Approach

The proposed system is a mobile application called myMove, designed to improve the parking experience for Malaysian drivers by providing clearer information, easier communication and more organized parking process. myMove focuses on giving drivers real-time parking availability with information, a reservation system, simple digital payment and a faster way to deal with double-parking.

The system allows users to view available parking spots in a building, select a space with details such as floor and parking number, and reserve it before entering. Once a spot is chosen, it becomes unavailable to others, reducing confusion and unnecessary searching around the same place. The app also sends reminders when the user's parking time is ending to help avoid extra charges. Additionally, a QR-based double-parking communication feature let blocked drivers scan a displayed QR Code to contact the car owner through in-app messaging or calls without revealing personal phone numbers.

2.5 Benchmark with Existing Systems

The comparative analysis of existing systems and the proposed myMove application is detailed in the table below:

TABLE 2.2: COMPARISON OF EXISTING SYSTEM

App Name	Functionality	Strengths	Weaknesses	Proposed Improvements
Touch 'n Go eWallet	It has parking features, such as LPR parking, street parking and provides travel passes, named My50 Pass.	Convenient and fast payment method while using LPR parking and reduces physical cash or card.	The balance in the physical card and eWallet balance lacks integration and support limited location only.	Aligns with myMove goal and expanding to more locations.
JomParking	Payment for street parking by providing city council's, allow by user by dragging the map and offer compound payment.	Convenient and time-saving features like paying for parking remotely and extending their time.	Supports only limited locations, provides only certain compound payment and only has online banking.	Aligns with myMove goal to show parking details and parking space selection
Flexi Parking	Users can pay for on-street parking via GPS, receive expiry reminders, extend parking remotely, and manage up to 10 vehicles in one account.	Supports over 46 municipal councils, monthly parking passes and prevents users from paying during non-chargeable hours.	App hard to navigate, buggy, errors when registration, provides certain locations and inaccuracy GPS	Aligns with myMove goal to show parking details and parking space selection

2.6 Summary

This chapter examined existing parking applications used in Malaysia to identify their features, strengths and weaknesses. Applications such as Touch ‘n Go eWallet Parking, JomParking and Flexi Parking were reviewed and found to mainly focus on digital payment while it lacks comprehensive parking information, provides limited coverage, minimal user interaction and lack of support for double parking situations. These findings justify the need for the proposed myMove application as more integrated and user-friendly parking solutions.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology that addresses the Objective 1 of myMove project, which focuses on examining the limitations of current car parking applications, studying existing parking solutions and identifying user needs before developing the proposed system. This structed research approach is adopted to ensure that the problems are clearly understood, and system requirements are correctly defined. For this project, the Waterfall Model is selected as the software development methodology because of its suitability for systems with well-defined requirements and a predictable development path. This model follows a linear structure in which each phase is completed thoroughly before moving on to the next part. The research design for Objective 1 includes a detailed literature review and a simple user survey to study existing parking applications, identify frustrations and analyze gaps in current system. Later phases including system design, development, usability testing, deployment and maintenance will also be planned, followed by the Waterfall methodology.

3.2 Proposed System Development Process

This project will be developed using the Waterfall Model, which follows step-by-step sequence from the initial study of requirements to the final maintenance of the system. The model is chosen for myMove because the requirements are clearly defined, stable, and suitable for a step-by-step development approach, allowing each phase to be carried out with clarity and minimal doubt. By completing one phase fully before moving to the next, the development process becomes more manageable, organized and easier to work with. Below shows how each stage of the Waterfall Model will be applied throughout the system development:

- Requirements Phase:
  - This phase is completed in FYP1, all information related to the parking problems, user needs, and system expectations is gathered and analyzed. This includes studying existing solutions, identifying limitations such as absence of real-time parking availability, lack of information and limited support for handling double parking situations. Feedback gathered through pilot study is used to define the functional and non-functional requirements that will guide system design and development in next phase.

- Design Phase:
    - The design phase focuses on turning the identified requirements it a clear system plan. This phase includes the creation of use-case diagram and UI prototypes to show how users will interact with the application. The UI designs developed in FYP1 prioritize simplicity and ease of use to demonstrate the main workflow, while more detailed and complete interface designs will be developed in FYP2.

- Development Phase:
    - The development phase will take place in FYP2, where the myMove application will be built based on the approved system design. Core features will be developed step by step, focusing on creating modular and maintainable code. This phase will also involve integrating the frontend and backend components to ensure smooth data flow and proper system functionality.

- Testing Phase:
    - Testing will be carried out throughout FYP2 to ensure the system works as expected and meets user requirements. Functional testing will be conducted to check that each feature operates correctly, while basic usability testing will be used to assess user interaction and interface clarity. Any issues or errors identified will be resolved before the system is deployed.

- Deployment Phase:
  - After development and testing are completed, the application will be deployed in a controlled environment for demonstration and evaluation. This phase allows the system to be reviewed as a fully working prototype and prepares for further improvement through user experience.

- Maintenance Phase:
  - The maintenance phase focuses on making improvements and refinements after the system is deployed. Feedback gathered during testing and evaluation will be used to fix issues, improve performance and enhance system features. This phase also considers system scalability and the addition of new functionality in future versions of the application.

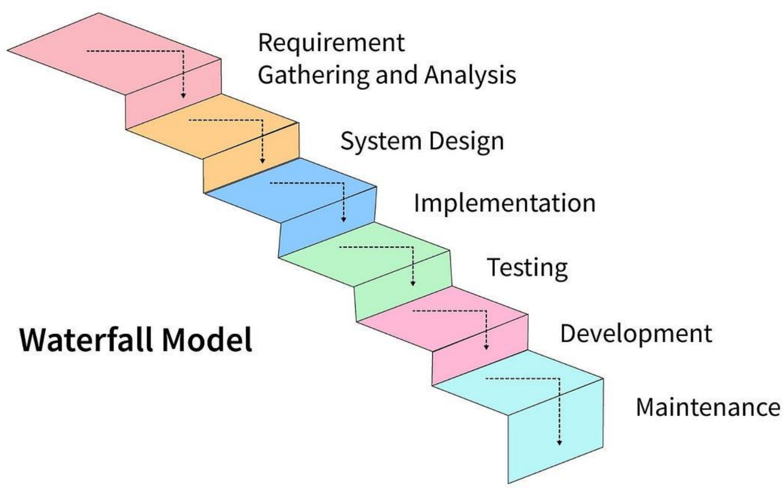

Figure 3.1 Waterfall Model that will be used for myMove project

3.3 Gantt Chart

Activity	September	October	November	December
Problem Statement & Topic Approval	✓			
Literature Review	✓	✓		
Pilot Study & Survey Data Collection		✓	✓	
Data Analysis & Findings			✓	
System Requirements			✓	✓
Use Case Diagram			✓	✓
UI Prototype Design			✓	✓
FYP1 Thesis Submission				✓

Figure 3.2 Gantt Chart of Project Timeline

3.4 Limitations and Features

To establish a foundation for myMove's development, the limitations of current car parking system and the desired features for the mobile application were identified through a literature review and pilot study findings.

3.4.1 Limitations of Current Systems

- **Lack of Mobile Accessibility:** Many malls and buildings rely on outdated systems that are not integrated with mobile applications, making it inconvenient for people to search for their parking or view how much time they spend.
- **Lack of Hardware Integration:** With parking availability relying on manual updates or no hardware system in place.
- **Missing Parking Info:** Current application doesn't show building name, floor and parking space number
- **Absence of Double-Parking Solutions:** Current systems do not provide any built-in mechanism to handle double-parking situations, leaving drivers without a secure and efficient way to contact other vehicle owners.
- **Lack of Integrated Communication Features:** Existing systems do not support in-app messaging and calling features for double-parking situations, forcing users to rely on external methods that may compromise privacy.

3.4.2 Proposed Features for myMove

Based on the pilot study with 44 people, that include university students and other individuals, the following features were identified as crucial for myMove to address these limitations:

- **Real-Time Parking Availability**: Allowing users to view available and occupied parking spaces in buildings and help drivers make faster parking decisions.
- **Parking Reservation**: Enabling users to reverse specific parking spaces in advance to reduce unnecessary searching.
- **Navigation Assistance**: Providing guided directions to the selected parking area to improve user and reduce travel time .
- **Parking Details:** Displays information such as parking fee rate, building name, floor level, parking space number and parking duration once the car is parked.
- **Emergency Mode:** Providing quick-access features that allow users to send urgent alerts in emergency situations.
- **Space for User Feedback:** Receive feedback from users / drivers about their experience in the specific place where they park so that we can improve.
- **QR Code-Based Solution for Double Parking:** Allow blocked drivers to scan a QR Code to contact car owners quickly without exposing confidential details.
- **In-App Messaging and VoIP Calling:** Enable secure communication between drivers through chat, and voice call within the application.

3.5 Data Collection Methods

To investigate the challenges faced by drivers and to gather user requirements for the parking system, the following methods were used:

Literature Review:

- **Source**: Existing mobile parking applications and publicly available documentation, including Touch ’n Go Parking, Flexi Parking, and JomParking, were reviewed to understand current practices and system capabilities.
- **Focus:** The review focused on identifying common limitations in existing systems, such as lack of real-time parking availability, poor user communication features, and limited integration between parking information and user interaction tools.
- **Process:** Information was gathered through feature analysis of existing applications, official app descriptions, user reviews, and observational evaluation conducted between 2020 and 2025. A comparative analysis was then performed to identify system gaps and areas for improvement.

Pilot Study:

- **Participants:** A total of 37 participants including university students, and working adults aged between 18 to 45 above have taken part in the online survey to share their experiences and consent about car parking systems
- **Method:** Google Form was used to collect responses on difficulties finding parking spaces, delays caused by blocked vehicles and user expectations for a smart parking application.
- **Data Collection**: The results indicate that most respondents had trouble in finding available parking spaces (56.8%) and encountered double parking situations (89.2%). Most participants expressed strong interest in features such as showing vehicle parking details (73%), digital payment integration (62.2%) and QR code that helps blocked drivers (51.4%) as a main feature.

3.6 Develop Module of myMove and Integration

While the development phase will take place in FYP2, this section outlines the approach for bringing together myMove core components based on the requirements gathered in Objective 1. myMove will be developed as a cross-platform mobile application using Flutter with Dart for the frontend. Although Flutter supports both Androids and iOS, the first release of myMove will be limited to Android devices only due to current hardware and incompatibility with Apple devices. The interface will be designed using Figma, focusing on clean and minimal layout and user-friendly experience.

The integration plan covers:

- **Frontend Development:**

Flutter will be used to develop an intuitive and responsive user interface compatible with Android devices. The frontend will include important key features such as:

- User login / registration (with secure authentication)
- Displays an interactive map for exploring nearby parking locations.
- A search bar for destinations and a bottom navigation to profile, QR scanner for double parking situation and history
- Shows a building-level parking layout with color-coded indicators for available, occupied, and reserved spaces.
- Users can select a parking spot, and the system updates its status when they arrive and extend parking duration
- Shows parking details (floor, section, lot, duration), lets users upload a photo, and provides fee info with instant payment.

- Backend Development:

The backend will be powered by Firebase that includes various product categories like authentications and more. The core backend responsibilities include:

  - Managing authentication (email, password, or Google sign-in) using Fireabase Authentication
  - Deliver instant push notification using Firebase Cloud Messaging (FCM)
  - Manage real-time parking status updates using Firebase Realtime Database
  - Support media uploads such as parking spot photos using Firebase Storage
  - Location based parking search and navigation guide using Google Maps API
  - Server-side processes such as payments and reservations using Cloud Functions.
  - Generate and scan QR codes using Flutter QR libraries (qr_flutter)
  - Communication and calls using Firebase Realtime DB and third-party API's
  - Payments will be handled using Stripe, Touch 'n Go eWallet, or any supported Malaysian gateway integrated through Firebase Functions.

- Database Structure:
  - Storing user details, building, floor and parking lot data for map and location retrieval using Firebase Firestore
  - Manage communication records (chat messaging and call) using Agora or Zegocloud

- Hardware Integration
  - Parking sensors will be installed in individual parking spaces to detect whether a spot is occupied or vacant using ESP32 microcontrollers.
  - Each ESP32 device will transmit parking status data wirelessly to the backend system through Wi-Fi and sync with Firebase Firestore.

3.7 Summary

This chapter describes the methodology used for the myMove project, where the Waterfall Model was applied to guide system development in a clear and structured way. It describes how user needs and system limitations were identified through a literature review and pilot study carried out during FYP1. The chapter also presents the proposed features and overall system, including frontend, backend, database and hardware. In addition, a Gantt chart is included to show the planned project timeline and development schedule.

CHAPTER 4

RESULT AND DISCUSSION

4.1 Introduction

This chapter presents the prototype outcomes of the proposed project myMove: Car Parking System, along with feedback gathered from a pilot study involving 37 participants including university students and working adults. The chapter begins by outlining the user interface (UI) design of myMove and the core system features developed. Following this, the chapter analyses the responses from users who participated in the pilot study. The purpose of this evaluation is to determine how well the application addresses in common parking challenges faced by drivers and how effectively it improves the overall parking experience compared to existing parking system like, Touch ‘n Go eWallet parking, Flexi Parking and JomParking. The insights obtained are essential for refining the systems design, enhancing usability and ensuring that the final application aligns with the needs and expectations of real users.

4.2 Result and Analysis of Pilot Study

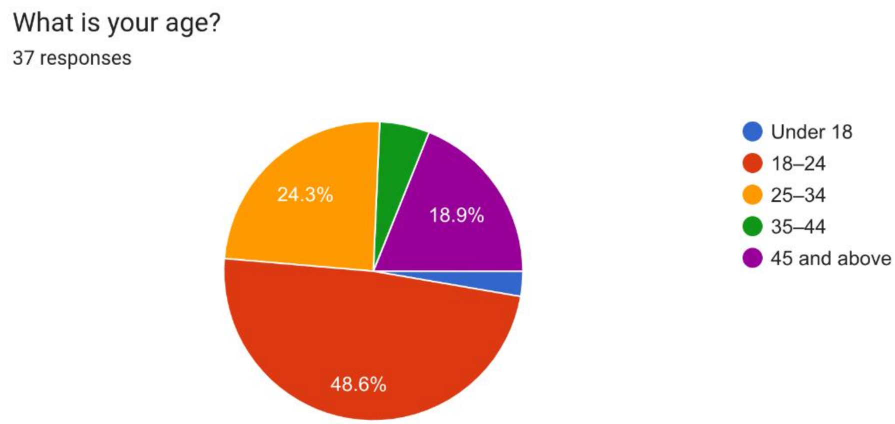

Figure 4.1 Age Distribution

The pie chart shows the ages of the people who participated in the pilot study. Almost half of them were young adults aged 18 to 24 with over 48.6%. The next highest group aged 25 to 34, making up 24.3 responses. The least number of responses were 45 and above with 18.9% and the smallest percentage are under 18 with 2.7%.

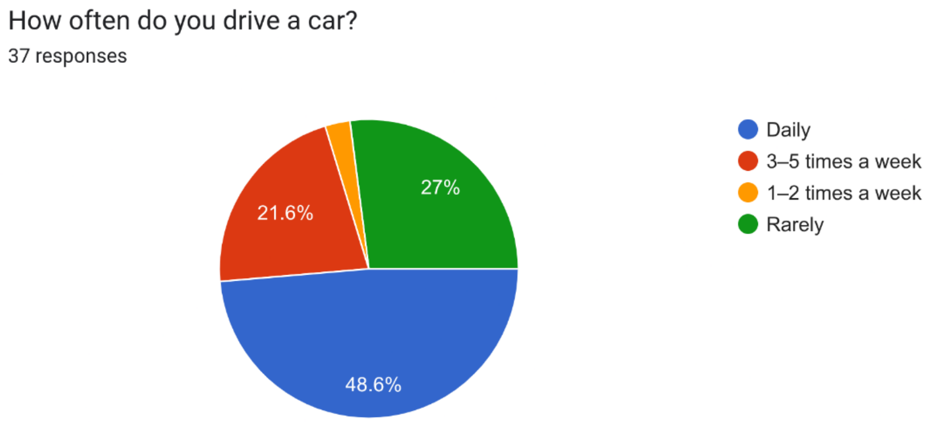

Figure 4.2 Frequency of Car Driving

The data shows how often the participants drive their car frequently. Almost half of them 48.6%, drive every day and another 21.6% drive 3 to 5 times a week. This means most participants are regular drivers. Meanwhile, 27% of respondents drive rarely and a small group of 2.7% drive 1 to 2 times a week.

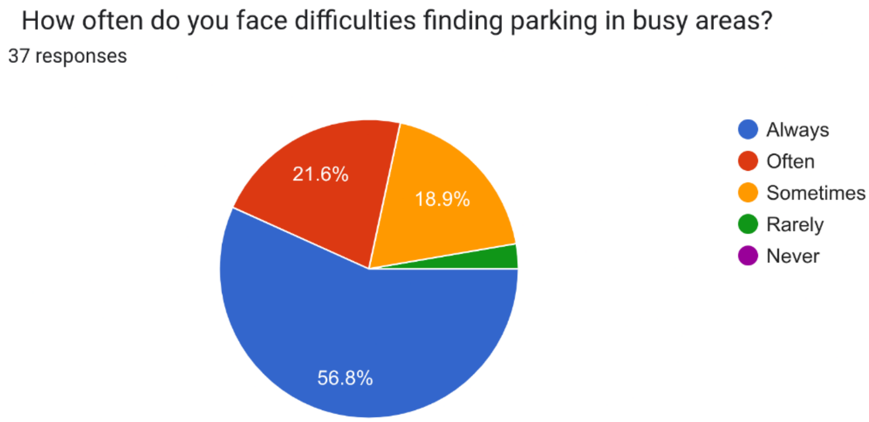

Figure 4.3 Frequency of Difficulties Finding Parking

The chart shows that many participants often have trouble finding parking in busy areas. 58.6% of participants said they always face this problem and another 21.6% said it happens often. This means about 78% of the respondents regularly struggle to find parking. Meanwhile, 18.9% of individuals said it happens sometimes, and small groups of people choose rarely, which is 2.7%.

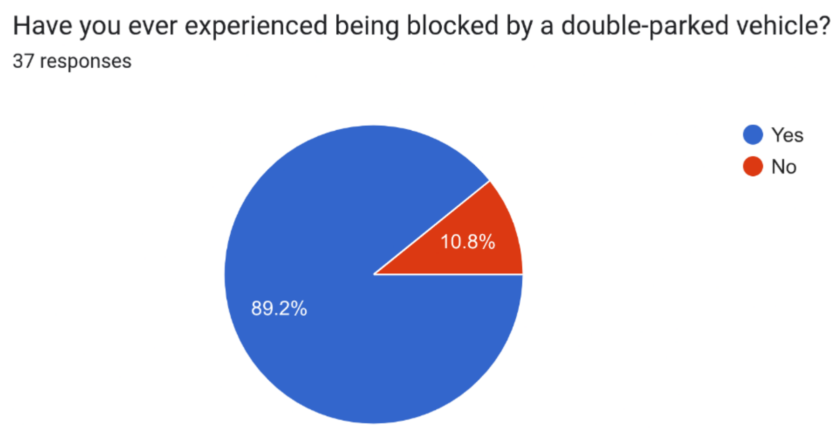

Figure 4.4 Experience with Double Parking Issue

The chart shows whether people have ever been blocked by a double-parked car. Almost everyone in the pilot study has faced this problem. A large majority of 89.2% said yes, they have been experienced with blocked car before. Only a small percentage of 10.8% said no. This clearly shows that being blocked by double-parked cars is a common problem for most drivers.

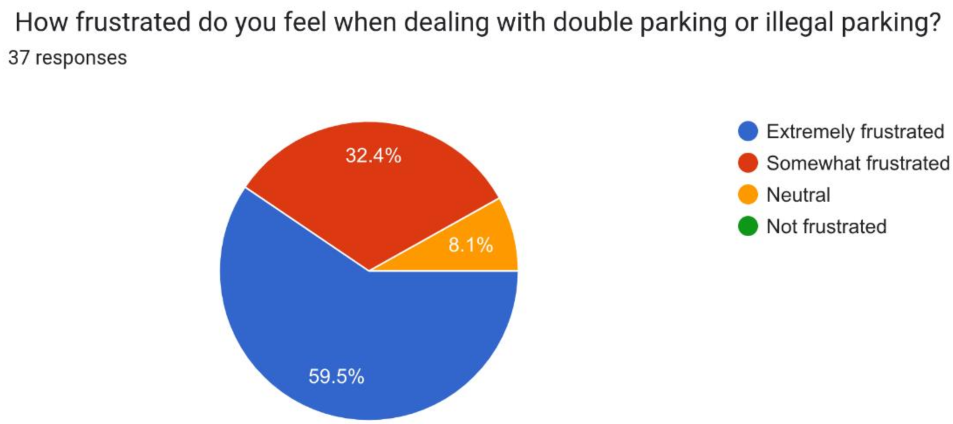

Figure 4.5 Level of Frustration Faced Due to Illegal Parking

This chart shows how frustrated people feel about double parking or illegal parking. Most people feel very upset about it. Nearly 60% (59.5% to be approx.), said they were extremely frustrated and another 32.4 were somewhat frustrated. This means more than 90% of the respondents feel annoyed by bad parking behavior. Only a small percentage of group, 8.1%, felt neutral.

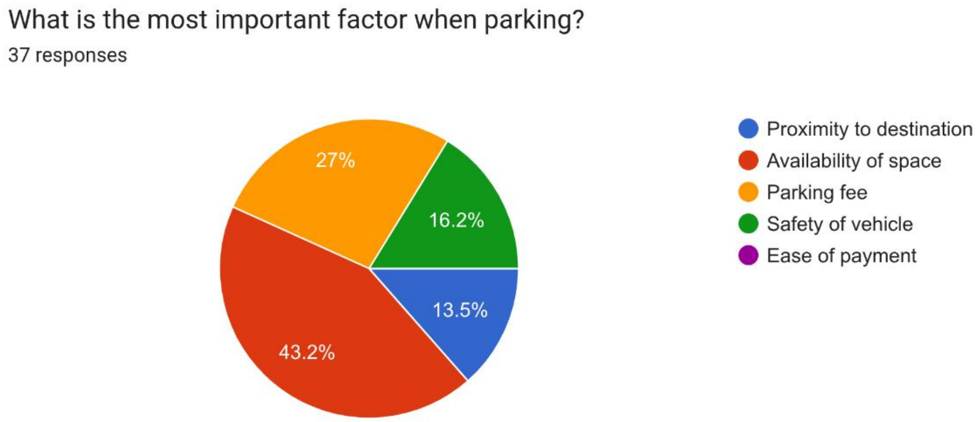

Figure 4.6 Most Important Factor When Parking

This chart shows what people care about the most when choosing a parking spot. The top priority is finding an available space, chosen by 43.2% of the respondents. The second most important factor is the parking fee, selected by 27%/ These two factors matter the most to the drivers. Vehicle safety comes at 16.2%, followed by how near the parking spot is to their destination at 13.5%.

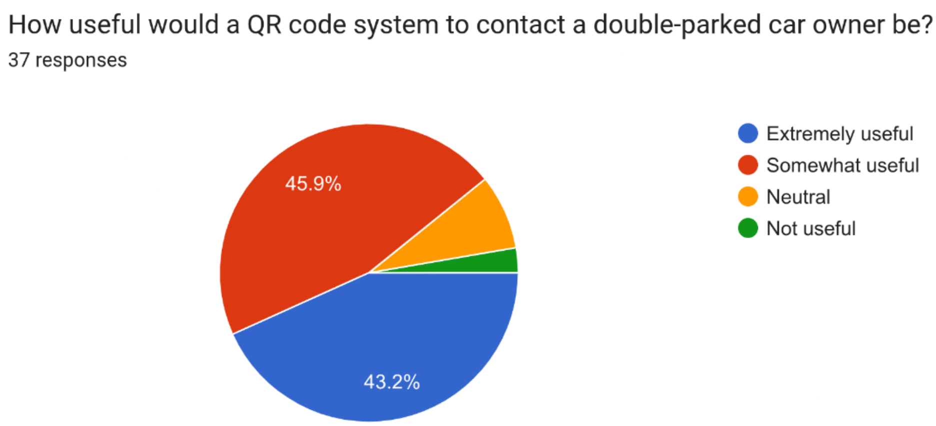

Figure 4.7 Useful of a QR Code System for Double Parking

This chart shows whether people think a QR code system for contacting a double-parked driver would be helpful. Almost everyone agreed that it would be useful. Nearly half of the respondents, which is 45.9%, said it would be somewhat useful, while 43.2% said it would be extremely useful. This means that almost 90% of the participants believe the system would help solve double-parking situations. Only a small group of individuals felt neutral with 8.1% and 2.7% said the system would not be useful.

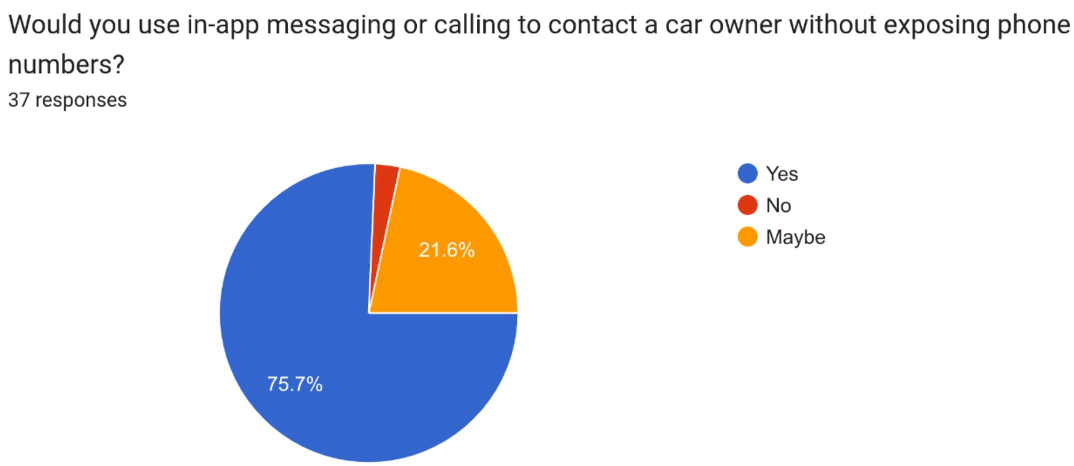

Figure 4.8 Willing to Use In-App Communication

This chart shows whether people use an app's messaging or calling feature to contact a car owner without sharing their phone number. Most participants, 75.7%, said yes, showing strong interest in a private way to communicate while another 21.6% said it maybe meaning they might use it depending on the situation. Only a very small percentage of people said no by 2.7%.

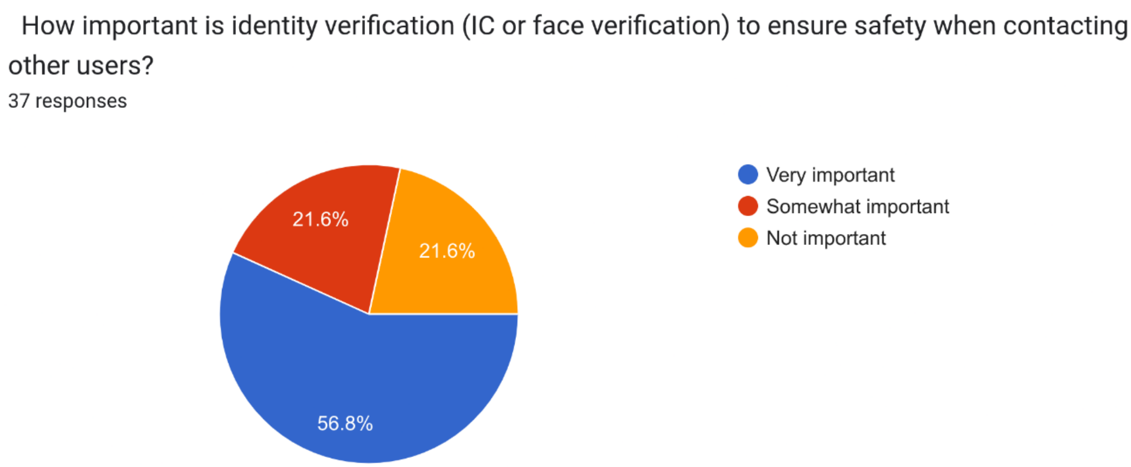

Figure 4.9 Important of Identify Verification for User Safety

This chart shows how important it is to identify verification using IC or face verification for user safety. Almost half of the respondents choose it very important by 56.8%, meanwhile 21.6% of individuals say it's somewhat important and another 21.6% said it's not important. Overall, most people prefer having verification because it makes the app safer.

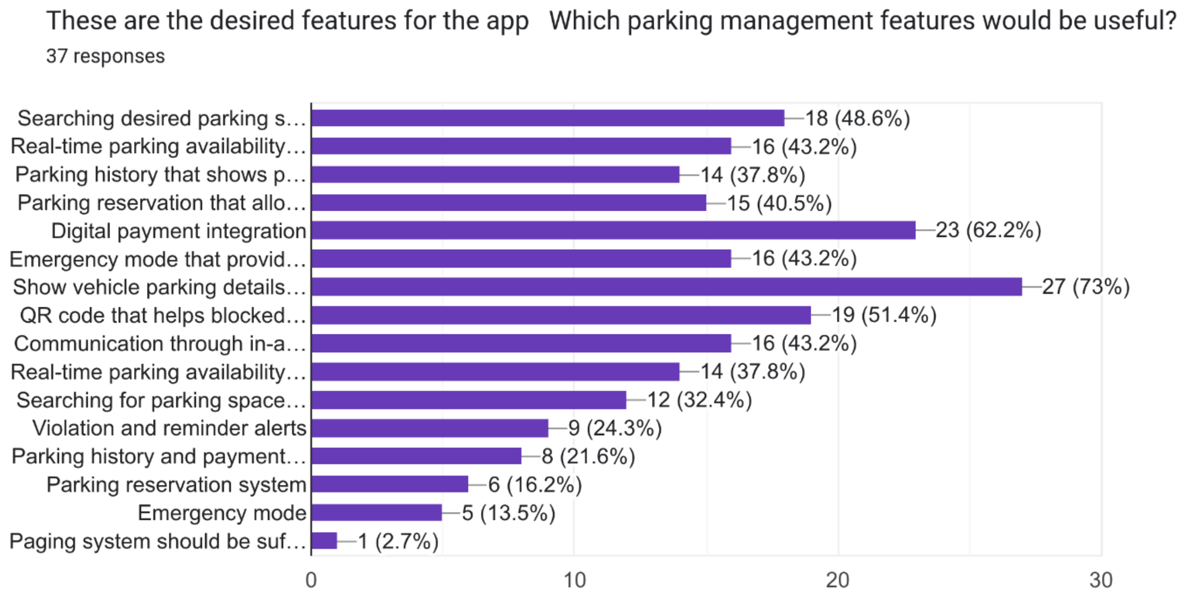

Figure 4.10 Desire Parking Features

The results clearly show which features users want the most in a parking system. The feature that people care about the most is the ability to see full vehicle parking details, chosen by 73% of the respondents. The second most important feature is digital payment integration (62.2%), showing that users want a quick and simple way to pay for parking. Many users also want tools that help with communication. About 51.4% liked the QR code system for contacting a blocked car, while 58.6% wanted a feature to search for parking spots. Real-time parking availability and in-app chat or VoIP calling were each selected by 43.2%, showing the need for better communication. In addition, 40.5% chose parking reservations, which shows that users want to plan before arriving. Overall, the findings show that apps must focus on providing clear parking information, clear parking information and communication for double parking issues, and smooth digital payments to meet user expectations.

4.3 Use Case Diagram

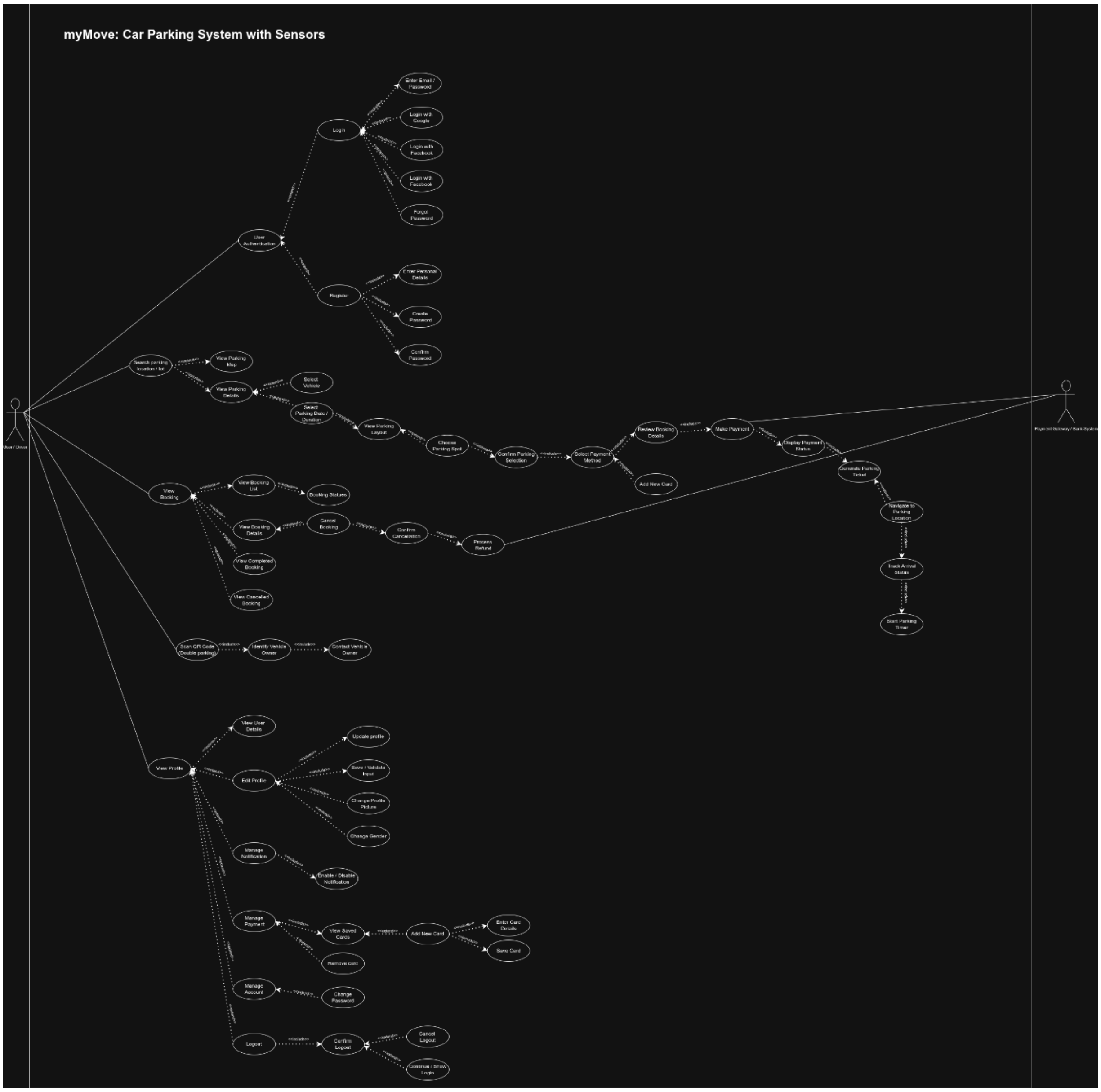

Figure 4.11 Use Case Diagram

4.4 UI Design

This describes the user interface (UI) design of the myMove application that was created using Figma, which emphasizes a clean layout, intuitive navigation and user-friendly interactions. The design aims to provide smooth and clear experience for drivers using the application in busy areas.

Key screen developed include:

Figure 4.12 Onboarding Screen

The onboard screen features the app logo and tagline to establish the identity of the app. A button is included to provide a clear entry point for users to begin using the app.

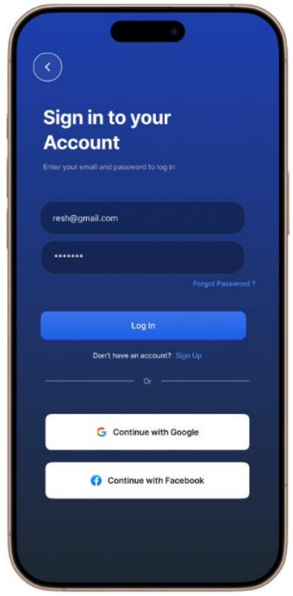

Figure 4.13 Login Page

Provides a login interface for existing users to access their accounts. It features email and password input fields, social login options and a clear path for new users to sign up.

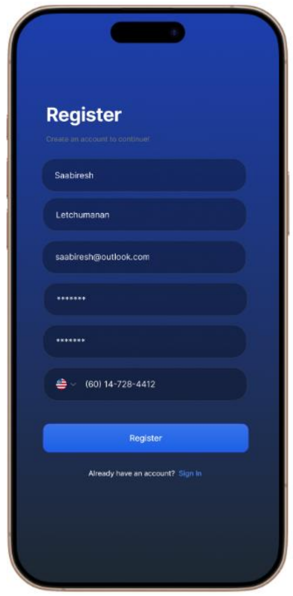

Figure 4.14 Register Page

The register page features input fields for user details and contact information to create a new account. A Register button submits the data, while a Sign In link allows existing users to return to the login page.

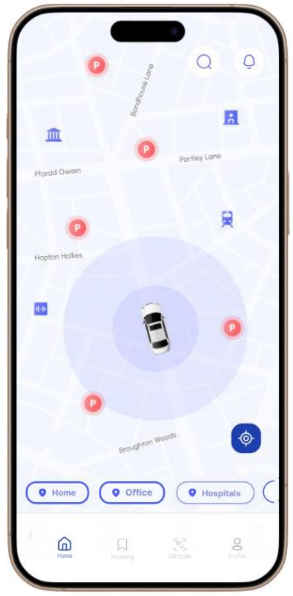

Figure 4.15 Home Page

This page displays an interactive map showing real-time parking locations and nearby landmarks to help users find available spots. It includes quick actions for common destinations and a bottom navigation bar.

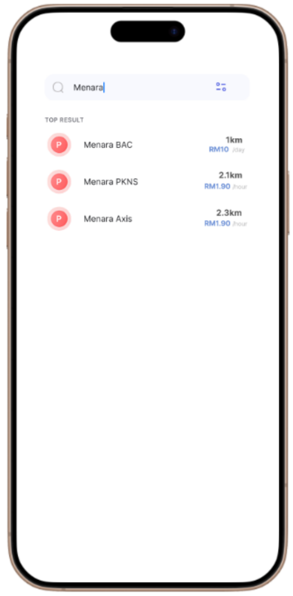

Figure 4.16 Search Parking

The search page allows users to find specific parking locations by name, displaying a list of top results with distance and pricing information. It has a search bar with filtering options to help users compare parking rates and proximity quickly.

Figure 4.17 Parking Details

This page displays comprehensive information about the parking facility, including its distance, operating hours and type such as "Underground". It also includes a descriptive overview and the total daily rate and a "Book Parking" button.

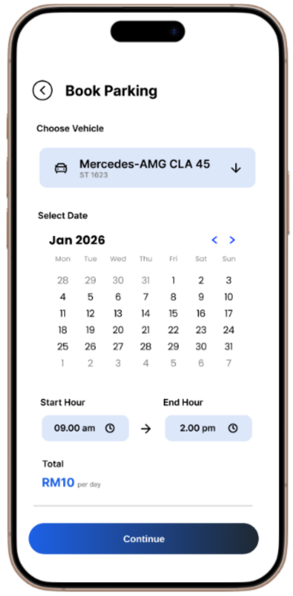

Figure 4.18 Book Parking

The book parking page allows users to select their registered vehicle and set a specific date using an interactive calendar. It features time pickers for start and end hours to calculate the total parking fee before proceeding with the reservation.

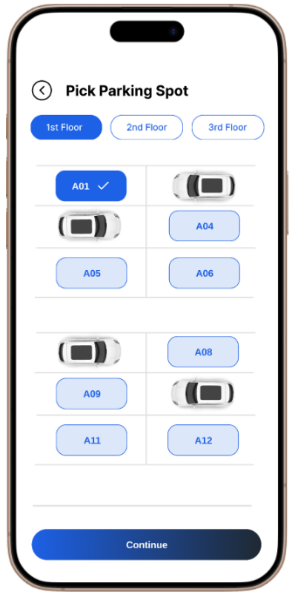

Figure 4.19 Parking Spot

The pick parking spot screen allows users to select a specific floor level and view an interactive layout of available bays. Users can choose a numbered spot, such as “A01” while occupied spaces are shown with car icons, making it easy to pick a spot.

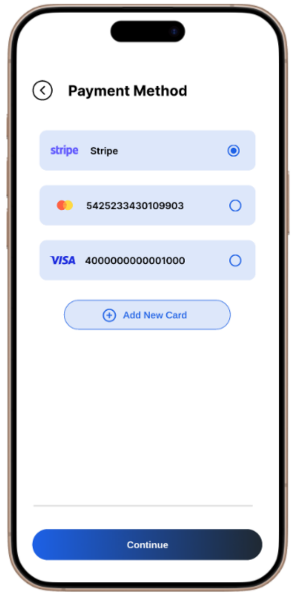

Figure 4.20 Payment Method

The payment method page allows users to select their preferred payment method, such as Stripe or a saved bank card. It also features an “Add New Card” button to facilitate quick and easy updates to the user’s financial details.

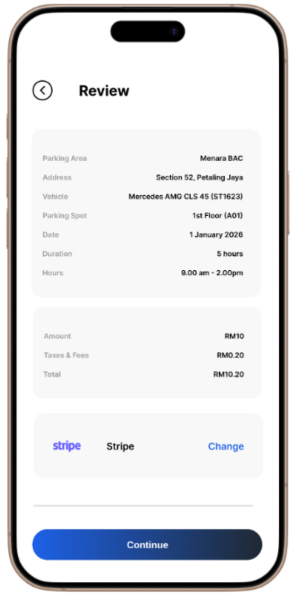

Figure 4.21 Review

The payment review page provides a comprehensive summary of the booking, including the parking area, specific spot, duration, and a detailed breakdown of the total costs. It allows users to verify their selected payment method and booking details before clicking "Continue" to finalize the transaction.

Figure 4.22 Navigation Assistance

The navigation page provides real-time, turn-by-turn directions from the user's current location to the selected parking facility, such as "Menara BAC." It features an interactive map with a highlighted route, a distance indicator, and a "Cancel" button to end the navigation session.

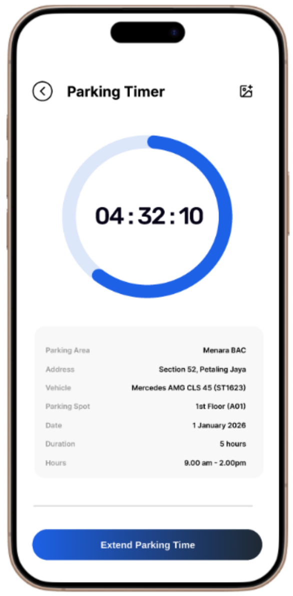

Figure 4.23 Parking Timer

Once the user passes the parking barrier, the parking timer page appears and shows a clear countdown ring and digital clock for the remaining parking time. It also displays the assigned parking spot and duration, with a button to easily extend the parking time.

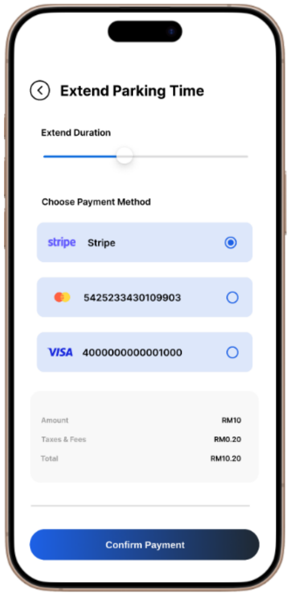

Figure 4.24 Extend Parking Time

This page lets users extend their current parking time by using a slider to choose the additional duration. It also shows the payment method, cost breakdown, and a “Confirm Payment” button to complete the extension.

4.5 Summary

This chapter summarized the results obtained during the FYP1 phase of the myMove project, based on the literature review and pilot study involving Malaysian drivers. The finding highlighted common parking-related problems such as difficulty finding available spaces, frequent double-parking incidents, and high levels of frustration among drivers. The proposed use case diagram and UI designs confirmed that the system's core functions are aligned with user needs, especially features related to view parking availability, reservation, parking information and digital payments. The survey results showed strong interest in a centralized parking management application, with high demand for clear parking details, integrated payment options, and a QR-based communication system. Overall, the results support the relevance of myMove and provide clear direction for FYP2, where full system development, implementation, and usability testing will be carried out.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS FOR FUTURE RESEARCH

5.1 Introduction

This chapter marks the conclusion of the first phase of the myMove research project. The focus of FYP1 was to examine the parking challenges faced by Malaysian drivers, including difficulties in finding parking spaces and issues caused by double parking, as well as to understand user expectation for a modern parking solution. The findings from this phase serve as a foundation for the development of the myMove platform in FYP2. This chapter summarizes the key outcomes of the study, discusses the contribution of the project, and provides recommendations for future research and system enhancement.

5.2 Summary of the Research Objectives

To achieve this a survey was conducted among 37 participants including university students and working adults using Google Forms. The analysis of the collected data revealed several common patterns. Many individuals reported have difficulties in current system when it comes to important factor when parking and being block by another vehicle. These findings provide a strong basis for designing a system that meets real user needs and help guide the development of FYP2.

5.3 Research Contributions

Although this study is part of FYP1, it still provides important contributions:

- Identified major issues faced by Malaysian drivers, including difficulty finding parking, lack of real-time parking information and frequent double-parking vehicle incidents from the user's perspective.
- Provided survey-based evidence showing demand for software-based parking management that combines all mentioned features and solutions.
- Defined the core functional requirements of the proposed system, such as parking space selection, real-time parking information, QR code-based solution for double parking, receive reminders and digital payment support.
- Established a clear and research-support foundation for further system design, development, and usability testing to be carried out in FYP2.

5.4 Practical Implications and Beneficiaries

The results of the pilot study are useful for multiple stakeholders:

- Drivers and Vehicle Owners: The proposed system addresses common parking issues by offering clear parking information, easier parking space selection and QR code-based solution for double parking, helping drivers save time.
- Property and Parking Operators: Provides guidance on creating user-friendly parking system that improve space usage, reduce congestion and increase management efficiency.
- Application Developer and Designer: The user feedback helps guide the design of simple interface and workflows that focus on real-world parking behaviors and user expectations.
- Future Researcher: The results provide a baseline for future studies on software-based parking management solutions and mentioned features in the project.

5.5 Limitations of the Present Study

The scope of this research in FYP1 was limited to the exploratory stage. The following constraints are noted:

- The survey involved 37 participants, which provided useful information but may not fully represent the experience of all drivers in Malaysia, including different age group and locations.
- Data was collected only through an online questionnaire, in-depth interviews or group discussions were not conducted, which limits deeper insights into user behavior.
- Since the application is still at the proposal and prototype stage, real-world testing and performance evaluation were not conducted.
- The study examined only a limited number of parking applications; it may not fully reflect the variety of solutions available.

5.6 Future Works

Future development will focus on several key enhancements to maximize the system's functionality and impact. The primary objective for FYP2 is the full implementation of the myMove mobile application, specifically the integration of UI design to frontend, backend and real-time data handling. Usability testing will also be expanded to a larger user base to precisely evaluate the user experience, system performance and overall performance.

Beyond the immediate scope of FYP2, future work aims to integrate smart parking hardware, such as sensors or barrier systems, to enable automatic detection of parking availability. The system could be further refined with advanced features like predictive parking suggestions, usage analytics and integration with city council or private parking providers. These enhancements are designed to transform the application into a scalable and intelligent parking system suitable for wider real-world deployment.

5.7 Summary

This chapter marks the completion of the first phase of the myMove project, which focuses on understanding parking-related challenges faced by drivers and identifying the need for a modern, software-based parking system. Through literature review and pilot survey, key issues such difficulty finding parking, lack of real-time information, stress caused by double parking and limited communication between drivers were identified.

The finding indicates strong user interest in a centralized and modern parking system that addresses common challenges faced by drivers in busy areas. Although the system has not been fully implemented or tested, the results provide a solid foundation for FYP2, where the complete application will be implemented, evaluated and refined based on user feedback.

REFERENCES

Zhen C., Z. Xu, Kang T., S. Jia (2025) *Environmental and Social Benefits of Urban Parking Space Shortages Mitigation Management Model: A System Dynamics and Nudge Approach*. 17(14), 6414–6414.

Group, A., & Group, A. (2024, December 23). *Efficient Parking Management Systems for Smart Cities - Abdul Rahman Al Shareef Group*. Abdul Rahman al Shareef Group.

Khalid, H. G., Abbas, H. H., Hussein, M. J., Mahmood Jawad Abu-AlShaeer, Mukhlif, S. K., & Dmytro Khlaponin. (2024). *Unveiling the Next Frontier: The Future of Connected Technologies*. 207–218.

Puspitasari, D., Noprianto, N., Afif Hendrawan, M., & Andrie Asmara, R. (2021). Development of smart parking system using internet of things concept. *Indonesian Journal of Electrical Engineering and Computer Science*, *24*(1), 611.

Kalašová, A., Čulík, K., Poliak, M., & Otahálová, Z. (2021). Smart Parking Applications and Its Efficiency. *Sustainability*, *13*(11), 6031.

Haslinda A., R. Ng, Rosmah M. (2016) *Factors Influencing Haphazard Parking in The Klang Valley, Malaysia 10.6007/IJARBSS/v6-i9/2303*

Anis.(2025) *KL Sees Over 1,000 Abandoned Cars in Just Five Months* https://www.carz.com.my/webview/2025/07/kl-sees-over-1000-abandoned-cars-in-just-five-months.

Aditya Kumar. (2022, December 20). *Has Technology Improved Our Lives? | Simplilearn*. Simplilearn.com; Aditya Kumar. https://www.simplilearn.com/how-has-technology-improved-our-lives-article

Najihah Mad Rosni, N., Fadzil Jobli, A., & Neging, M. (2019). Driver Parking Behaviour: An Observational and Experimental Intervention Study. *IOP Conference Series: Materials Science and Engineering*, *601*(1), 012002.

Sampath, L. P., & Boggavarapu, K. S. S. (2025). *IoT-based Smart Parking System for Industry 4.0 with QR code access and real-time navigation. EPJ Web of Conferences*, *336*, 03003.

Siti Aslinda Mahat, & Maslinda Mohd Nadzir. (2025). *ASSESSING USER SATISFACTION OF SMART PARKING PAYMENT SYSTEM IN SELANGOR*. *Journal of Digital System Development*, *3*(1), 13–28.

Lim, B. (2017, July 31). *Uni students develop smart system to curb double parking problems*. NST Online; New Straits Times. https://www.nst.com.my/lifestyle/bots/2017/07/262578/uni-students-develop-smart-system-curb-double-parking-problems